

Analysis 2: HW5

Aamod Varma

March 2, 2026

Problem 1. We have $M = B(0,1)$, $F : \mathbb{R}^3 \rightarrow \mathbb{R}^3$ a smooth vector field. And n an outward unit vector on $\partial M = S^2$ the boundary of M .

1. We have our collection of open sets $\cup_i B_i$, now note that using result from prev hw, first we have a collection of functions f_j for $j \in J$ such that for each f_j we have some i for which $\text{supp} f_j \subseteq B_i$ and that we have $\sum_{j \in J} f_j = 1$. Note that these functions are locally finite in that for each $x \in M$ there are only a finite amount of function f that are non-zero at that point. Now simply define $F_j = F f_j$. Since both F and f_j are smooth their product is smooth. And note we now have $\sum F_j = \sum F f_j = F \sum f_j = F$

2. We have,

$$\int \int \int_{\mathbb{R}_+^3} \text{div} G = \int \int \int_{\mathbb{R}_+^3} \partial_x G_1 + \partial_y G_2 + \partial_z G_3$$

Now note that we can consider the triple integral individually for the three functions above. When we do this as G has a compact support this means that the improper integral $\int_{-\infty}^{\infty} \partial_x G_1 = 0$ and $\int_{-\infty}^{\infty} \partial_y G_2 = 0$ simply by using FTC. More specifically as the support is compact it's bounded hence when we consider the improper integral we'll be well outside the support region where G_1, G_2, G_3 will all be zero. Hence using FTC since we have $G_i(b) - G_i(a) = \int_b^a \partial_i G_i$ we end up getting $0 - 0 = 0$ (note the abuse of notation here, G is technically $\mathbb{R}^3 \rightarrow \mathbb{R}$, but above we're using a, b as scalars although they should be vectors, in this case as we go to infinity from any direction we'll end up outside the compact support so it doesn't matter). So we end up rewrite our equation as,

$$\int \int \int_{\mathbb{R}_+^3} \text{div} G = \int \int \int \partial_z G_3$$

Now we can apply FTC again on the right side and note we have $\int_0^{\infty} \partial_z G_3 = G_3(x, y, \infty) - G_3(x, y, 0)$. Again since G has compact support $G_3(x, y, \infty) = 0$ and hence we have $\int_0^{\infty} \partial_z G_3 = -G_3(x, y, 0)$. But note that $-G_3$ is simply $G \cdot (-e_3)$ and as $z = 0$ and we're integrating over x, y we're essentially integrating over $\partial \mathbb{R}_+^3$ and hence we get,

$$\begin{aligned} \int \int \int_{\mathbb{R}_+^3} \text{div} G &= \int \int \int \partial_z G_3 \\ &= \int \int -G_3(x, y, 0) \, dx \, dy \\ &= \int \int_{\partial \mathbb{R}_+^3} G \cdot (-e_3) \, dS \end{aligned}$$

3. Using spherical coordinates we have,

$$\begin{aligned} \iiint_M \text{div} F &= \int_0^{\pi} \int_0^{2\pi} \int_0^1 (\text{div} F)(\text{Jacobian}) \, dr \, d\theta \, d\phi \\ \iiint_M \text{div} F &= \int_0^{\pi} \int_0^{2\pi} \int_0^1 \partial_r(r^2 F_r \sin \theta) + \partial_{\theta}(r \sin \theta F_{\theta}) + \partial_{\phi}(r F_{\phi}) \, dr \, d\theta \, d\phi \end{aligned}$$

Now we know that $F = \sum F_j$, so we have $F_r = (F \circ W) \partial_R W = ((\sum F_j) \circ W) \partial_r W = \sum (F_j \circ W) \partial_r W$. If we define $F_{rj} = (F_j \circ W) \partial_r W$ then we have $F_r = \sum F_{rj}$. Now do the same for both θ and ϕ . So this gives us,

$$\iiint_M \operatorname{div} F = \int_0^\pi \int_0^1 \int_0^{2\pi} \partial_r \left(r^2 \sum F_{rj} \sin \theta \right) + \partial_\theta \left(r \sin \theta \sum F_{\theta j} \right) + \partial_\phi \left(r \sum F_{\phi j} \right) dr d\theta d\phi$$

Now consider each of the terms separately, starting with θ we have,

$$\int_0^\pi \int_0^1 \int_0^{2\pi} \partial_\theta \left(r \sin \theta \sum F_{\theta j} \right) d\theta dr d\phi$$

Now note that as F_j has a compact support, so does $F_{\theta j} = (F_j \circ W) \partial_r W$. This means that in the interval $0 \rightarrow 2\pi$ we have some a, b such that the function F_j is identically zero at a and b , or in other words the compact support is defined from $a \rightarrow b$ and the function is zero outside this. So if we consider each of the terms in the sum separately i.e for an arbitrary F_j note that,

$$\int_0^\pi \int_0^1 \int_0^{2\pi} \partial_\theta (r \sin \theta F_{\theta j}) d\theta dr d\phi = \int_0^\pi \int_0^1 r \int_a^b \partial_\theta (\sin \theta F_{\theta j}) d\theta dr d\phi$$

Now we can use FTC to write $\sin \theta F_{\theta j}(b) - \sin \theta F_{\theta j}(a) = \int_a^b \partial_\theta (\sin \theta F_{\theta j}) d\theta$. But at a, b $F_{\theta j}$ is identically zero as we showed above. Hence the LHS is just zero. So the integral evaluates to zero. So we have,

$$\begin{aligned} \int_0^\pi \int_0^1 \int_0^{2\pi} \partial_\theta (r \sin \theta F_{\theta j}) d\theta dr d\phi &= \int_0^\pi \int_0^1 r \int_a^b \partial_\theta (\sin \theta F_{\theta j}) d\theta dr d\phi \\ &= \int_0^\pi \int_0^1 r \cdot 0 = 0 \end{aligned}$$

Now as each arbitrary $F_{\theta j}$ go to zero, the sum goes to zero. And we get,

$$\int_0^\pi \int_0^1 \int_0^{2\pi} \partial_\theta \left(r \sin \theta \sum F_{\theta j} \right) d\theta dr d\phi = 0$$

Similarly for ϕ we have,

$$\int_0^1 \int_0^{2\pi} \int_0^\pi \partial_\phi \left(r \sum F_{\phi j} \right) d\phi d\theta dr = 0$$

However, now note that for r from 0, 1, the function F_{rj} need not go to zero at the endpoints necessarily. So when we're considering F_{rj} then note that for arbitrary j we have $(r^2 F_{rj})(1) - (r^2 F_{rj})(0) = \int_0^1 \partial_r (r^2 F_{rj} \sin \theta) dr$ but here the LHS evaluates to $F_{rj}(1)$, so overall we get,

$$\begin{aligned} \int_0^{2\pi} \int_0^\pi \int_0^1 \partial_r \left(r^2 \sum F_{rj} \right) d\phi d\theta dr &= \int_0^{2\pi} \int_0^\pi \sum F_{rj}(1) dr d\phi d\theta \\ &= \int_0^{2\pi} \int_0^\pi F_r(1, \theta, \phi) dr d\phi d\theta \\ &= \int_0^{2\pi} \int_0^\pi F(W(1, \theta, \phi)) \partial_r W dr d\phi d\theta \end{aligned}$$

But now note that the inputs to F are with $r = 1$ which is the boundary of our ball i.e the 2-D sphere. So this is essentially evaluating F at the boundary. So as long as we consider just the boundary with radius 1 we can just evaluate F directly. In addition note that $|\partial_r W| = |(\sin \theta \cos \theta, \sin \theta, \sin \phi, \cos \theta)| = 1$ so the outward normal vector is basically just $\partial_r W$. Hence we end up with,

$$\iiint_M \operatorname{div} F = \iint_{\partial M} (F \cdot n) dS$$

Problem 2.

1. Firstly we know that f' is continuous and that $f'(p)$ is invertible. Now note that we know that the determinant of a matrix is a continuous function - which means that $\det f'$ is a continuous function as well. Now if $f'(p)$ is invertible then this means $\det f'(p) = k \neq 0$. Now choose $\varepsilon = \frac{|k|}{2} < |k|$, then we have some δ such that,

$$|\det f'(p) - \det f'(x)| < \varepsilon \quad \text{when} \quad |x - p| < \delta$$

But note this gives us,

$$\begin{aligned} |\det f'(p) - \varepsilon| &< |\det f'(x)| \\ \frac{k}{2} &< |\det f'(x)| \end{aligned}$$

Note that this means $\det f'(x) \neq 0$ or that f' is invertible in the δ -neighborhood around p .

Now let $A = f'(p)$, and define $g(p) = A^{-1}f(p)$. Note that A is a matrix and is unique for each point p . Then at p note that $g'(p) = A^{-1}f'(p) = A^{-1}A = I$. Now as f' is continuous and as in our δ neighborhood we have invertible matrices, so can approximate the matrices $f'(x)$ with A within some error. This gives us,

$$\|g'(x) - I\| < \varepsilon/1000$$

Now WLOG, take $p = 0$ (note we can do this because all we're doing is translating the graph of f so that the point we're considering is at 0, but linear translations and so on doesn't affect the invariability) so we have $f(0) = 0, f'(0) = Id$ or that $g(0) = A^{-1}f(0) = 0$, so we can do (basically from problem 3 of the exam)

$$\begin{aligned} g(p+h) - g(p) &= \int_0^1 g'(th)h \, dt \\ g(h) - h &= \int_0^1 g'(th)h - h \, dt \\ |g(h) - h| &\leq \int_0^1 |g'(th)h - h| \, dt \leq |h|\varepsilon \end{aligned}$$

Using triangle inequality this gives us,

$$1 - \varepsilon < \frac{|g(h)|}{|h|} < 1 + \varepsilon$$

Now consider x, y close to point $p = 0$ and using FTC we get,

$$\begin{aligned} g(x) - g(y) &= \int_0^1 g'(y + t(x-y))(x-y) \, dt \\ (g(x) - g(y)) - (x-y) &= \int_0^1 g'(y + t(x-y))(x-y) - (x-y) \, dt \\ &\leq \varepsilon(x-y) \end{aligned}$$

Using the same reasoning again we can get,

$$|g(x) - g(y)| \geq (1 - \varepsilon)|x - y|$$

which means that $g(x) \neq g(y)$ or that we get injectivity for $x \neq y$.

Now using part 2 of q3 of the exam we get some small radius for which we have,

$$B(0, (1 - \varepsilon)r) \subset g(U) \subset B(0, (1 + \varepsilon)r)$$

where U is the open ball with radius r . Note that this is basically a scaled version of the exam problem, i.e scaled down using r . Now note that this gives us surjectivity as for any y in the open

ball around $B(0, (1 - \varepsilon)r)$ that y is in $g(U)$ and hence there is some x for which $g(x) = y$. So we have surjectivity and hence bijectivity which means we have an inverse g^{-1} .

However, note that $g = A^{-1}f$ so $Ag = f$ and $(Ag)^{-1} = g^{-1}A^{-1} = f^{-1}$. So we just showed there is some open set around that point p given that the derivative has an inverse at p where f has an inverse.

2. Now we need to show that $g = (f)^{-1}$ is C^k . Clearly we have $g(f(x)) = x$, now we'll get a candidate function for g' which in this case would look like $g'(f(x))f'(x) = 1$ so $g'(f(x)) = f'^{-1}(x)$. Now we need to show that this is a frechet derivative. So let $x' = f(x)$ then we have $f(x+h) = x' + h'$, so now we then have $g(x' + h') - g(x') = h$, now as f' is differential we have $h' = f'(x)h + \text{remainder}$ so h would be $f'^{-1}(x)(h' - \text{remainder})$. Now note that we then have $f'^{-1}h' = h + \text{remainder}f'^{-1}(x)$, so we get, $|g(x' + h') - g(x') - f'^{-1}h'| = f'^{-1}(x)\text{remainder}$ but as h goes to zero we have the remainder go to zero as well by definition, so we have g has a frechet derivative. Now we have g' . We can do induction and now use f''^{-1} to get a derivative for g' i.e g'' . As f is C^k we can simply do this k times to get g to be C^k as well.

Problem 3. We have the differential form

$$\omega = f_1(x)dx_2 \wedge dx_3 - f_2(x)dx_1 \wedge dx_3 + f_3(x)dx_1 \wedge dx_2$$

We first do partition of unity on F to get $F = \sum F_j$ where F_j is supported a compact subset of B_j where $\cup B_j$ is an open cover of M . Now we need to show that $\int_{\partial M} \omega = \int_{\partial M} (F \cdot n) dS$. First we have the parametrization of the boundary using T i.e,

$$T(y_1, y_2) = (x_1(y_1, y_2), x_2(y_1, y_2), x_3(y_1, y_2))$$

and we know that,

$$n = \frac{\partial_1 T \times \partial_2 T}{|\partial_1 T \times \partial_2 T|}$$

Now when we use the parametrization T we have $dS = |\partial_1 T \times \partial_2 T| dy_1 dy_2$ so essentially we get,

$$\int_{\partial M} (F \cdot n) dS = \int_{\partial M} F(T(y_1, y_2)) \cdot (\partial_1 T \times \partial_2 T) dy_1 dy_2$$

Now using the formula for $u \cdot (v \times w)$ this is equal to,

$$\begin{aligned} \int_{\partial M} F(T(y_1, y_2)) \cdot (\partial_1 T \times \partial_2 T) dy_1 dy_2 &= \int_{\partial M} f_1(\partial_1 x_2 \partial_2 x_3 - \partial_1 x_3 \partial_2 x_2) \\ &\quad - \int_{\partial M} f_2(\partial_1 x_1 \partial_2 x_3 - \partial_1 x_3 \partial_2 x_1) \\ &\quad + \int_{\partial M} f_3(\partial_1 x_1 \partial_2 x_2 - \partial_1 x_2 \partial_2 x_1) \end{aligned}$$

However, now note that $dx_i \wedge dx_j = \partial_1 x_i \partial_2 x_j - \partial_1 x_j \partial_2 x_i$, so we basically have,

$$\int_{\partial M} F(T(y_1, y_2)) \cdot (\partial_1 T \times \partial_2 T) dy_1 dy_2 = \int_{\partial M} \omega$$

putting everything together we get,

$$\int_M (F \cdot n) dS = \int_{\partial M} \omega$$

Now for the next equality, apply the exterior derivative on ω , for that we first apply it on some f_i , so we have,

$$df_i = \frac{\partial f_i}{\partial x_1} dx_1 + \frac{\partial f_i}{\partial x_2} dx_2 + \frac{\partial f_i}{\partial x_3} dx_3$$

Now note that for df_1 if we consider $df_1(x)(dx_2 \wedge dx_3)$, then most terms will vanish as $dx_i \wedge dx_i = 0$ so we get,

$$df_1(x)(dx_2 \wedge dx_3) = \frac{\partial f_1}{\partial x_1} dx_1 \wedge dx_2 \wedge dx_3$$

doing this for each f_i we have,

$$d\omega = \left(\frac{\partial f_1}{\partial x_1} + \frac{\partial f_2}{\partial x_2} + \frac{\partial f_3}{\partial x_3} \right) dx_1 \wedge dx_2 \wedge dx_3$$

So we have,

$$\int_M d\omega = \int_M \left(\frac{\partial f_1}{\partial x_1} + \frac{\partial f_2}{\partial x_2} + \frac{\partial f_3}{\partial x_3} \right) dx_1 \wedge dx_2 \wedge dx_3$$

But the right side is basically,

$$\int_M \operatorname{div} F$$

so we have,

$$\int_M d\omega = \int_M \operatorname{div} F$$